

TESLA CHALLENGE "AIRFRAME" · BUSINESS MEETS TECH 2026

Airframe explained: why factory Wi-Fi fails and nobody notices

Ten figures, explained from scratch. Every number comes from the **eight captures** Tesla gave us. Read them in order and you will know what the problem is and what we are supposed to build.

8 sensors 30 minutes, 16 Sep 2026, 12:24–12:54 UTC 29 Access Points 60 devices
1,118,853 radio frames

THE SEQUENCE

Fig. 1 Why a Wi-Fi dead spot costs money

Fig. 3 Eight sensors, eight channels

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Fig. 8 A real case over 30 minutes

Fig. 10 The task: from haystack to alert

Fig. 2 Who is on the air in the hall

Fig. 4 A sensor only sees the envelope

Fig. 6 How a device connects

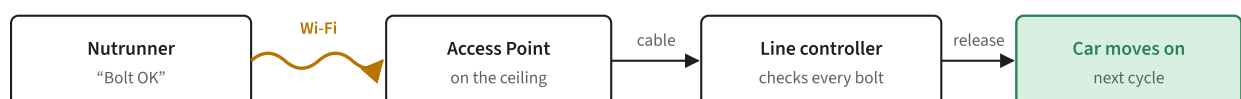
Table 1 Five patterns we look for

Fig. 9 Why we have to combine the sensors

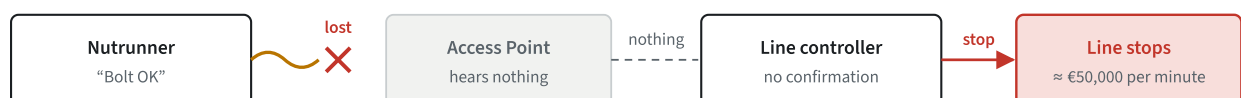
FIG. 1 Why a Wi-Fi dead spot costs money

On the line, a worker tightens bolts with a cordless nutrunner. The tool reports over Wi-Fi whether each bolt is seated correctly. Only then does the controller release the car.

Normal



Wi-Fi drops

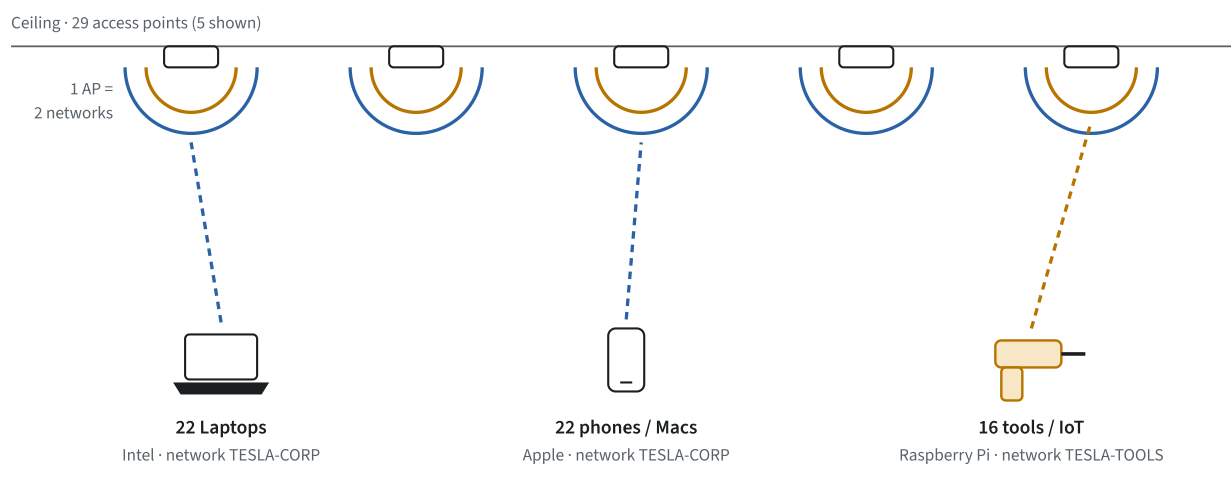


According to Tesla, **a car worth about €50,000 leaves the plant every minute**. If a tool loses its radio link, the line can stop even though the tool itself works fine.

KEY POINT Wi-Fi is part of production, not just an office convenience. A dead spot costs money.

FIG. 2 Who is on the air in the hall

Access points (APs), the Wi-Fi transmitters, hang from the ceiling. Each AP broadcasts **two networks** at once: one for tools and one for laptops and phones. On the floor are the devices trying to connect.



TESLA-TOOLS · network for tools

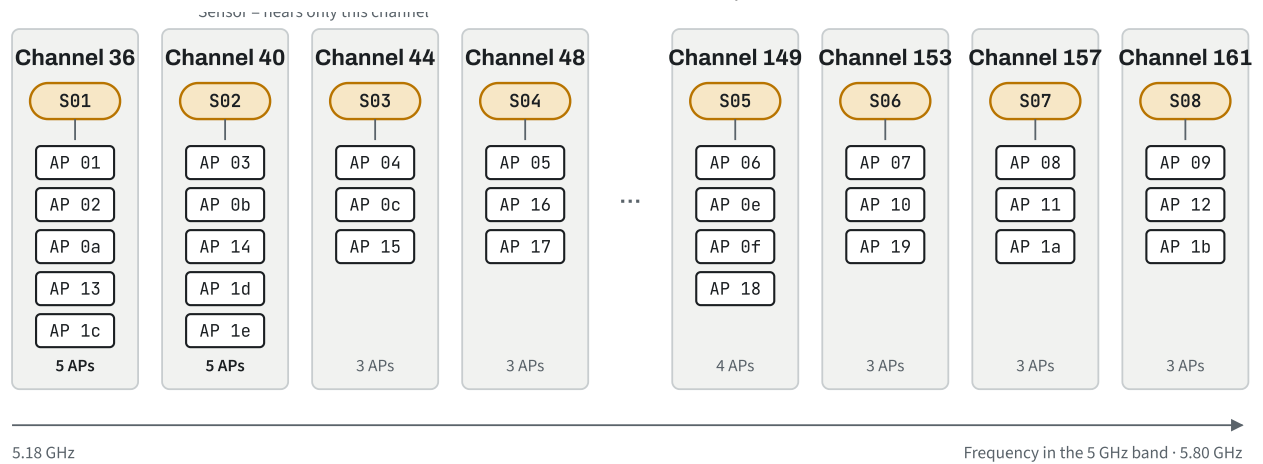
TESLA-CORP · network for laptops and phones

The first digits of a device address (MAC address) reveal the manufacturer. That the Raspberry Pis are tools or sensors is an **assumption**. What is certain: they only connect to TESLA-TOOLS.

KEY POINT Two networks of different importance. A tool without a connection is worse than a phone without one.

FIG. 3 Eight sensors, eight channels

Wi-Fi transmits on different **channels**, like radio stations on different frequencies. Each AP uses exactly one channel. Tesla set up eight sensors, and each one listens to **only one channel**.



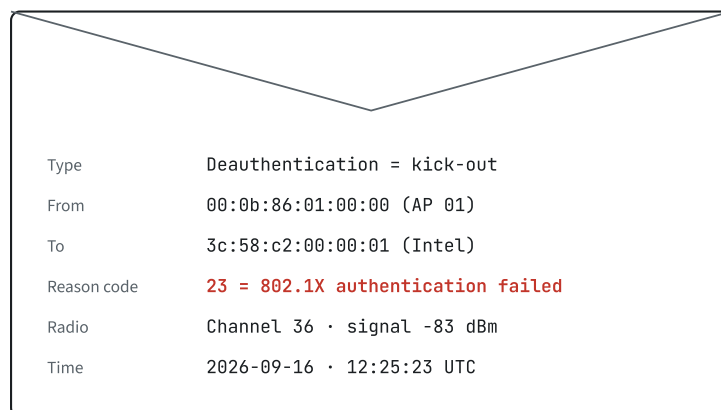
Each channel has its own sensor (S01–S08). The columns list the actual AP numbers from the captures. On channels 36 and 40, **five APs** share the same frequency; on the others, three or four.

KEY POINT A sensor is like a radio locked to one station. It does not hear what happens on other channels.

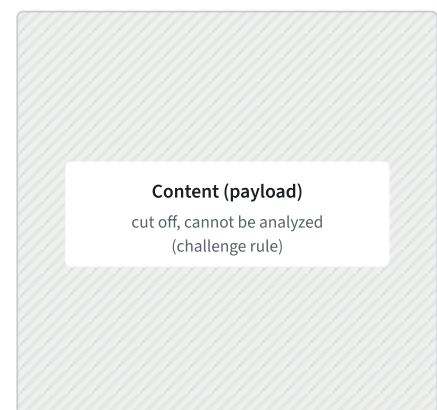
FIG. 4 A sensor only sees the envelope

Every radio frame is like a letter. The sensors only record the **envelope** (the “header”: who to whom, what kind of message, when, how loud). The content is cut off; that is a rule of the challenge.

Visible: the envelope (header)



Invisible: the letter

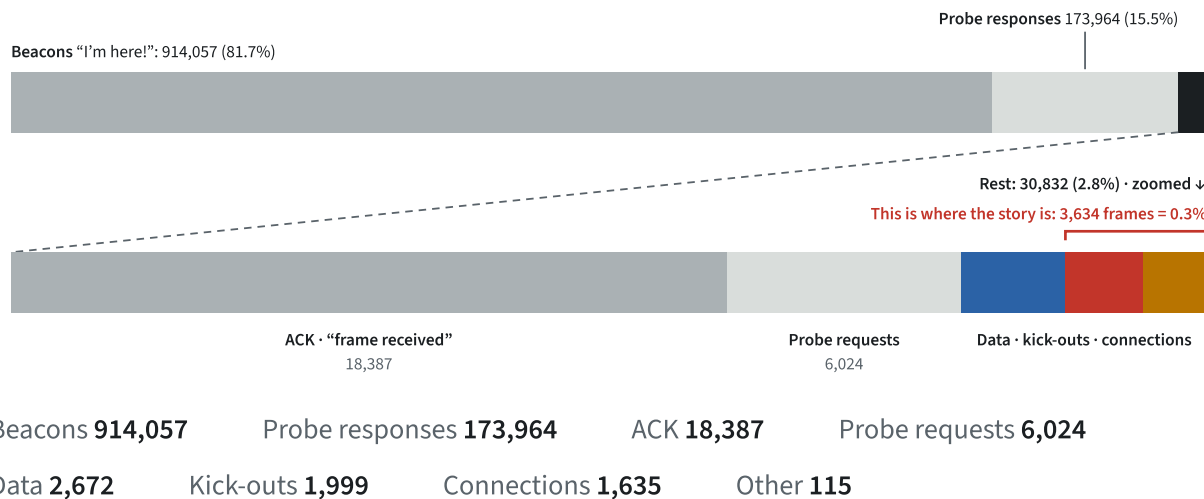


A real frame from sensor01.pcap. The envelope alone says a lot: access point 01 kicks out an Intel device because its authentication failed. **-83 dBm** is the signal strength at the sensor. The closer to 0, the stronger; -83 is weak.

KEY POINT We do not see *what* is said, only *who sends what kind of message to whom and when*. That is enough to find problems.

FIG. 5 The needle in the haystack

In 30 minutes the eight sensors record over a million frames. Almost all of it is routine: APs constantly calling out “I’m here!”. The interesting frames are a tiny fraction.

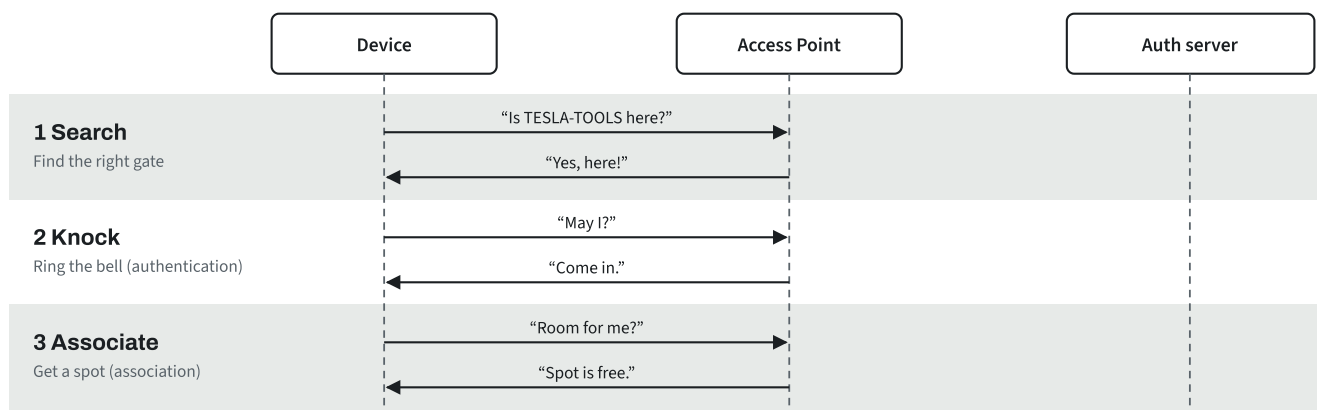


Beacons are each network’s “I’m here!”, about ten times per second. **Probe requests and responses** are devices searching for networks and the APs’ replies. **ACK** means “frame received”. Kick-outs cover deauthentication and disassociation; connections cover authentication and association.

KEY POINT The first job is filtering. Out of 1.1 million frames, we need the few thousand that tell a story.

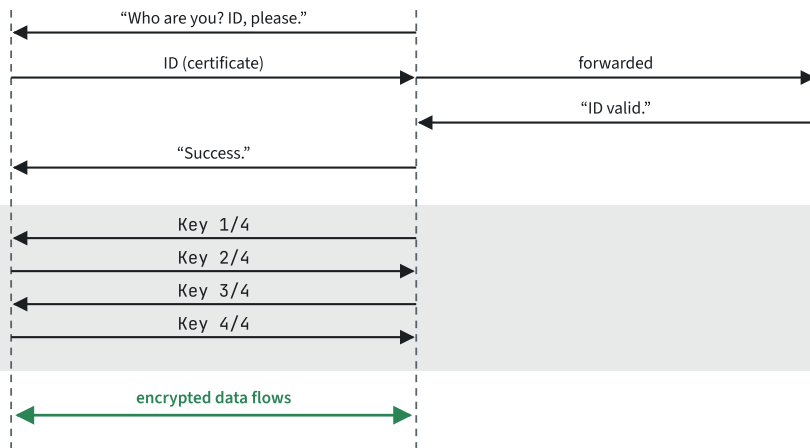
FIG. 6 How a device connects

Before a device may send data, it goes through five steps, like an employee at the factory gate. Every arrow is a frame the sensors can see.



4 Show ID

802.1X / EAP
Gatekeeper checks
with security

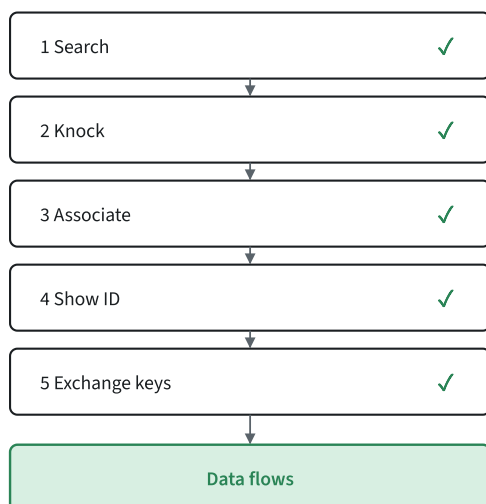
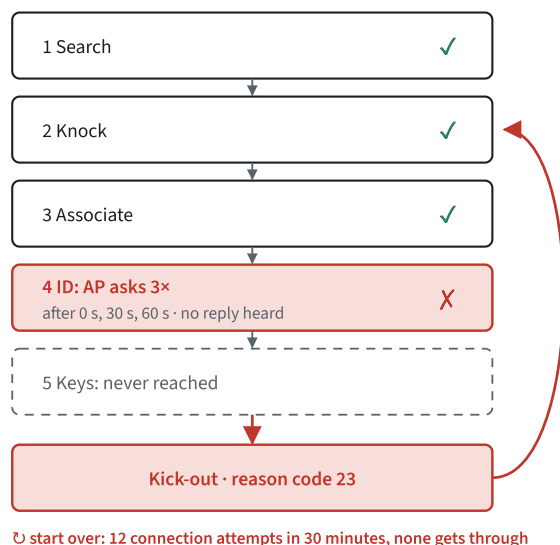
**Done**

The **authentication server** (technically “RADIUS”) sits somewhere in the data center. The AP asks it whether the ID is valid. That check runs over cable, so the sensors do **not** see it. They only see the radio side between device and AP.

KEY POINT A successful connection has a fixed sequence of frames. That sequence is our yardstick.

FIG. 7 A failure is a missing step

Nowhere in the data does it say “error”. You spot a problem when the sequence from Fig. 6 **breaks off**. Tesla puts it this way: “Failures hide as breaks in the 802.11 / 802.1X sequence, not as a single obvious frame.”

How it should work**Real case from sensor 01**

On the right is device 3c:58:c2:00:00:01 on AP 01. The sensor hears “ID, please” three times, then the kick-out. Whether the device never replies or the sensor just cannot hear the reply is impossible to tell from one sensor alone (see Fig. 9).

KEY POINT We are not looking for error messages. We are looking for incomplete sequences.

TABLE 1 Five patterns we look for

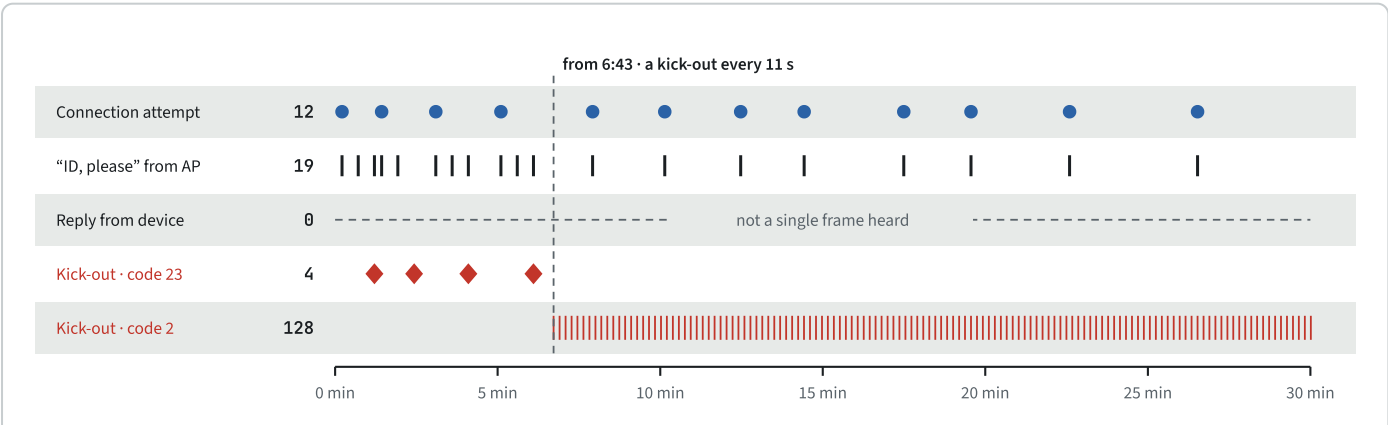
Tesla does not say which faults are hidden and only reveals them at the end. They typically fall into these categories:

PATTERN	WHAT IT LOOKS LIKE IN THE DATA	EFFECT IN THE HALL
Auth loop	Step 4 or 5 never completes, kick-out with code 23, retry, again and again	Device never gets online
Deauth storm	Many kick-outs in a short time, often at a fixed interval or to many devices at once	Misconfiguration or an attack with spoofed kick-outs
Roaming problems	Device keeps bouncing between APs or sticks to an AP with a weak signal	Dropouts when workers or vehicles move
Congestion	Many APs on one channel, many retries, missing acknowledgements	Slow, connection stalls
Missing or rogue APs	An AP goes silent, a network is missing, an unknown transmitter calls itself TESLA	Dead spot or security risk

First observations, still **unconfirmed**: number 13 is missing from the AP numbering (1 to 30), and some APs broadcast only one of the two networks.

FIG. 8 A real case over 30 minutes

The same device as in Fig. 7, now over the whole capture. Each mark is a frame sensor 01 heard. This is what a problem looks like on a timeline.



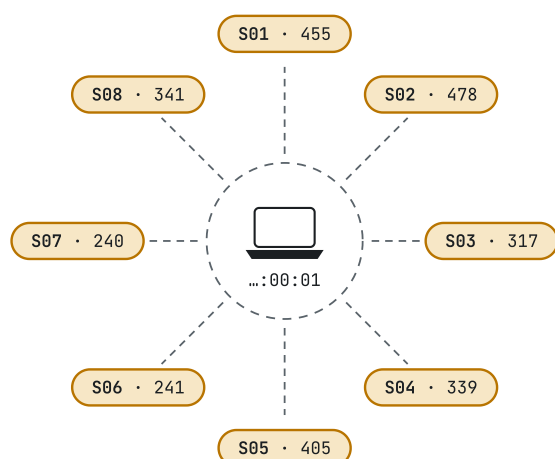
First the device fails authentication four times (code 23). From 6:43 onward the AP kicks it out **every 11 seconds** until the capture ends: 129 times, with code 2 (“previous authentication no longer valid”). Such a regular rhythm is suspicious. It looks more like automation or misconfiguration than chance.

KEY POINT Single frames say little. Only the **timeline per device** makes a problem visible.

FIG. 9 Why we have to combine the sensors

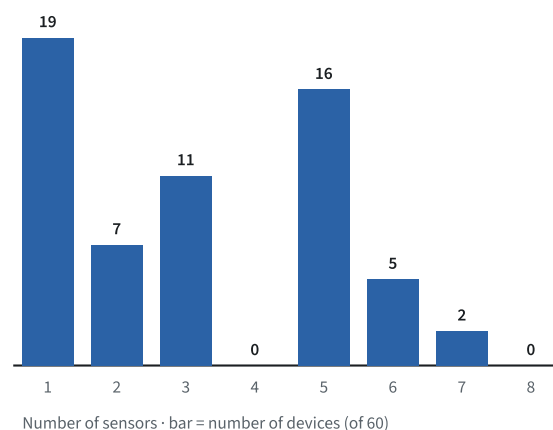
Devices search for networks on all channels, so the same device shows up at several sensors. That is both an opportunity and a trap: more sensors give more clues, but they also count the same event several times.

One device, eight viewpoints



Number = AP replies to this device heard · own frames: 0

How many sensors hear a device?

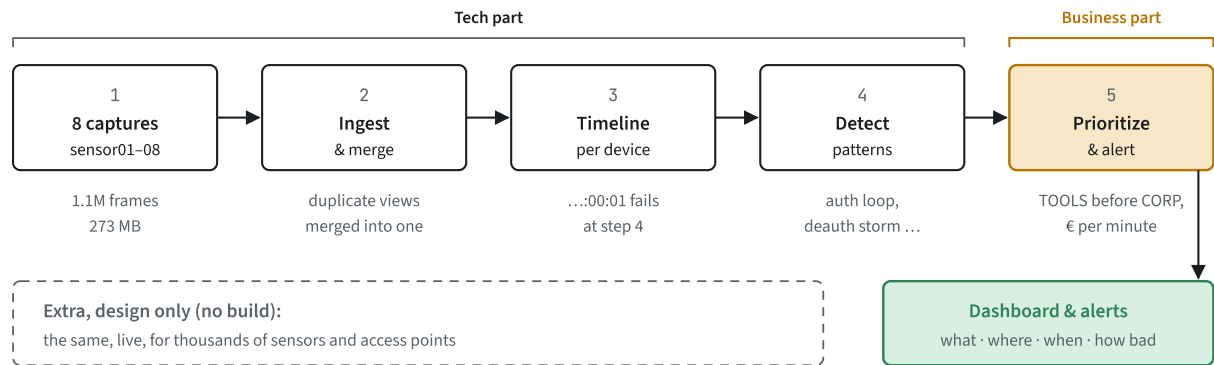


Left: no sensor hears a single frame from the device in Fig. 8. Yet all eight know it exists, because APs on every channel reply to it. So it is reconstructed from **indirect traces**. **Right:** most of the 60 devices that transmit are heard by several sensors. These views must be merged into **one** timeline.

KEY POINT One sensor alone sees only a slice. The full picture only appears when you merge the eight views.

FIG. 10 The task: from haystack to alert

Tesla expects a working prototype that reads the captures, finds problems, uses more than one sensor and shows the findings on a dashboard, plus a short pitch.



Stages 1 to 4 are engineering. **Stage 5 is our edge:** not every fault matters equally. A tool on TESLA-TOOLS that cannot connect can stop the line; a phone on TESLA-CORP cannot. Per the brief, the design for thousands of sensors is optional and does not need to be built.

KEY POINT The goal is not “yet another Wi-Fi monitor”. It is one that knows **which fault costs production money.**

GLOSSARY The terms at a glance

Access Point (AP)

Wi-Fi transmitter on the ceiling. There are 29 here.

SSID

Name of a network, here TESLA-TOOLS and TESLA-CORP.

Frame

A single radio message.

Beacon

A network’s “I’m here!”, about ten times per second.

Deauthentication

Kick-out, always with a reason code (e.g. 23, 2).

RADIUS

The authentication server that checks the ID.
Invisible to the sensors

Client / device

Anything that wants to connect: laptop, phone, tool.

Channel

Radio frequency. Each sensor hears exactly one.

Header

The “envelope” of a frame: sender, receiver, type, time.

Probe

Searching for networks (“Is ... here?”) and the reply.

802.1X / EAP

Showing ID: authentication with a certificate or password.

4-Way-Handshake

Four messages in which device and AP agree on the encryption key

dBm

Signal strength. -40 is very strong, -80 weak, -90 barely usable.

pcap

File format for captures, readable e.g. with Wireshark.

Data: sensor01.pcap to sensor08.pcap, 2026-09-16, 12:24–12:54 UTC. Our own analysis of the headers with a Python parser. Manufacturers are derived from MAC prefixes (3c:58:c2 Intel, f0:18:98 Apple, b8:27:eb Raspberry Pi). The “tool” role for the Raspberry Pis is an assumption. According to Tesla, the captures contain hidden faults that are only revealed at the end. Figs. 7 and 8 show a suspicious pattern, not a confirmed finding.

